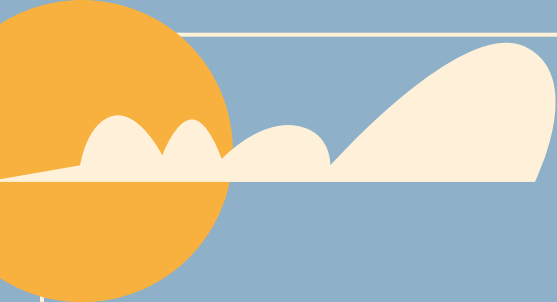


SkyWatch Charlotte

Flight Delay Prediction

Anshuk Gottipati, Nicholas Cassarino,
Sania Khan, Purva Jagtap, Frank Garcia





6th

Busiest airport in the world

58.8 million

Passengers in 2024

596,583

Arrivals and departures in 2024



Source: Charlotte-Douglas Official Website

Delays are bad for everyone

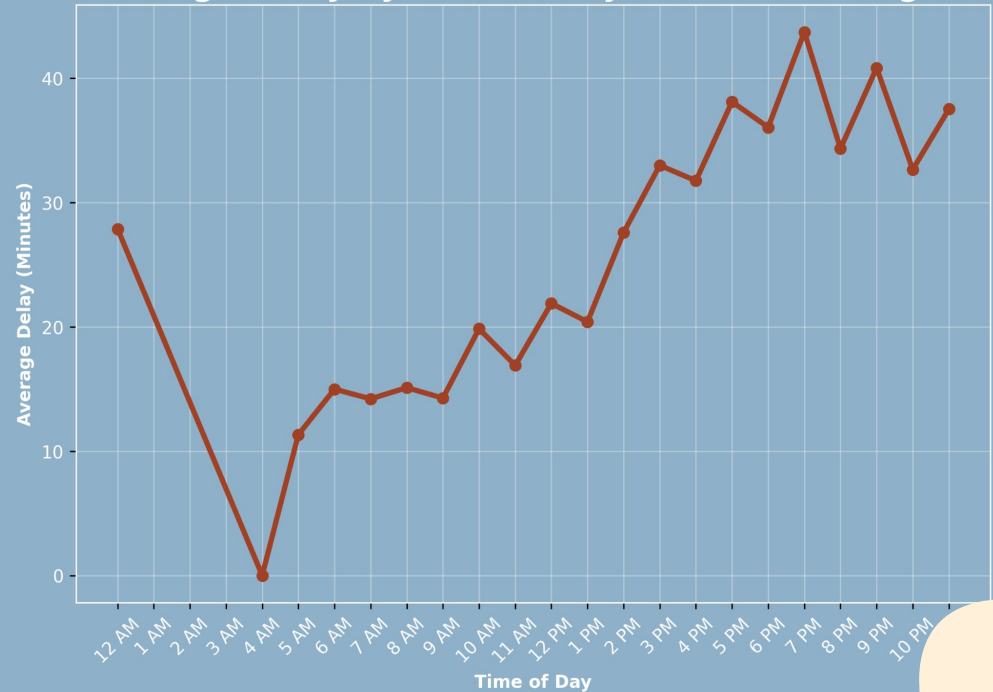
Delays cause passengers:

- Stress
- Missed connections
- Wasted money
- Lost time

Delays cost airports & airlines

- Customers
- Business Partners

Average Delay by Hour of Day — Charlotte Flights



Source: US DOT Bureau of Transportation

What causes delays?



Carrier Delay

Could be due to cleaning, loading, late crew, inspection, fueling, etc.



Late Aircraft

Delay of the next flight due to a late arrival of the same aircraft earlier.



Weather

Extreme or hazardous weather conditions at point of departure, arrival, or enroute.

Weather Delays

Patterns

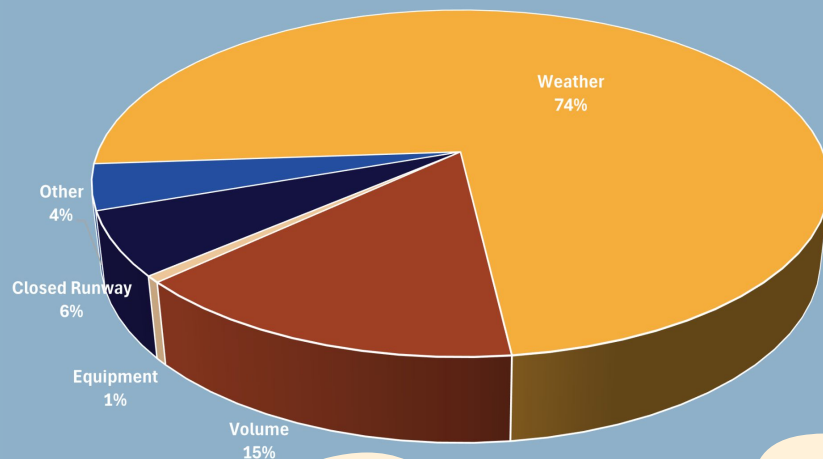
The type of weather that causes delays and when in the year is highly variable from year to year.

Capacity

If an airport is running near capacity, its ability to handle weather delays worsen.

For airports like KCLT, it can sometimes take hours to catch up to demand when weather causes delays.

OPSNET Standard Reports, Delay by Cause, June 2017 - May 2023



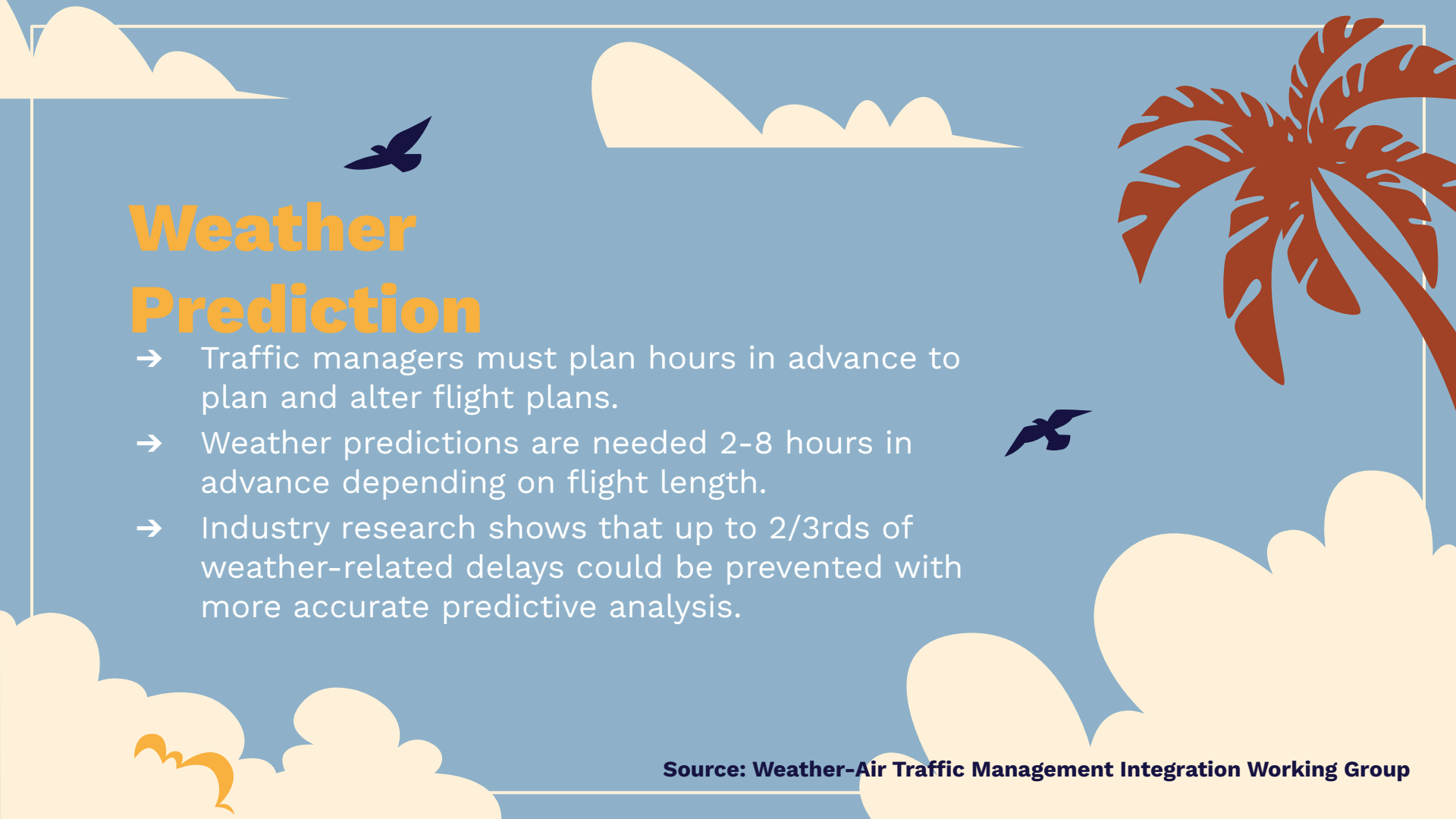
Source: Federal Aviation Administration



\$30,000,000,000

Lost annually by U.S. airports and airlines from weather delays

Source: Joint Economic Committee Majority Staff



Weather Prediction

- Traffic managers must plan hours in advance to plan and alter flight plans.
- Weather predictions are needed 2-8 hours in advance depending on flight length.
- Industry research shows that up to 2/3rds of weather-related delays could be prevented with more accurate predictive analysis.

Source: Weather-Air Traffic Management Integration Working Group

Our Solution

Using a combination of publicly available flight data and the same weather observations airports use, create an application that predicts the likelihood of flight delays at Charlotte-Douglas and an interface that anyone can use to access it.





Application Workflow



Data Ingestion

Take the US Department of Transit's flight data and combine it with raw weather observations from the airport

Pre-Process

Cleaning data, parsing weather observations, and creating features.

Machine Learning

Run combined flight and weather data through our machine learning model to predict if there will be any delay.

Stream

Stream results to the user in an accessible and readable format.

User Interface

Allows passengers to easily look up their flight and see if a delay is likely

SkyWatch
CLT AirportPredictorAbout

Flight Delay Predictor

Select a destination, date, and flight to get an AI-powered delay prediction.

Check Your Flight

Enter your details to analyze delay probability

Destination

Select destination ▾

Departure Date

📅 Select valid date

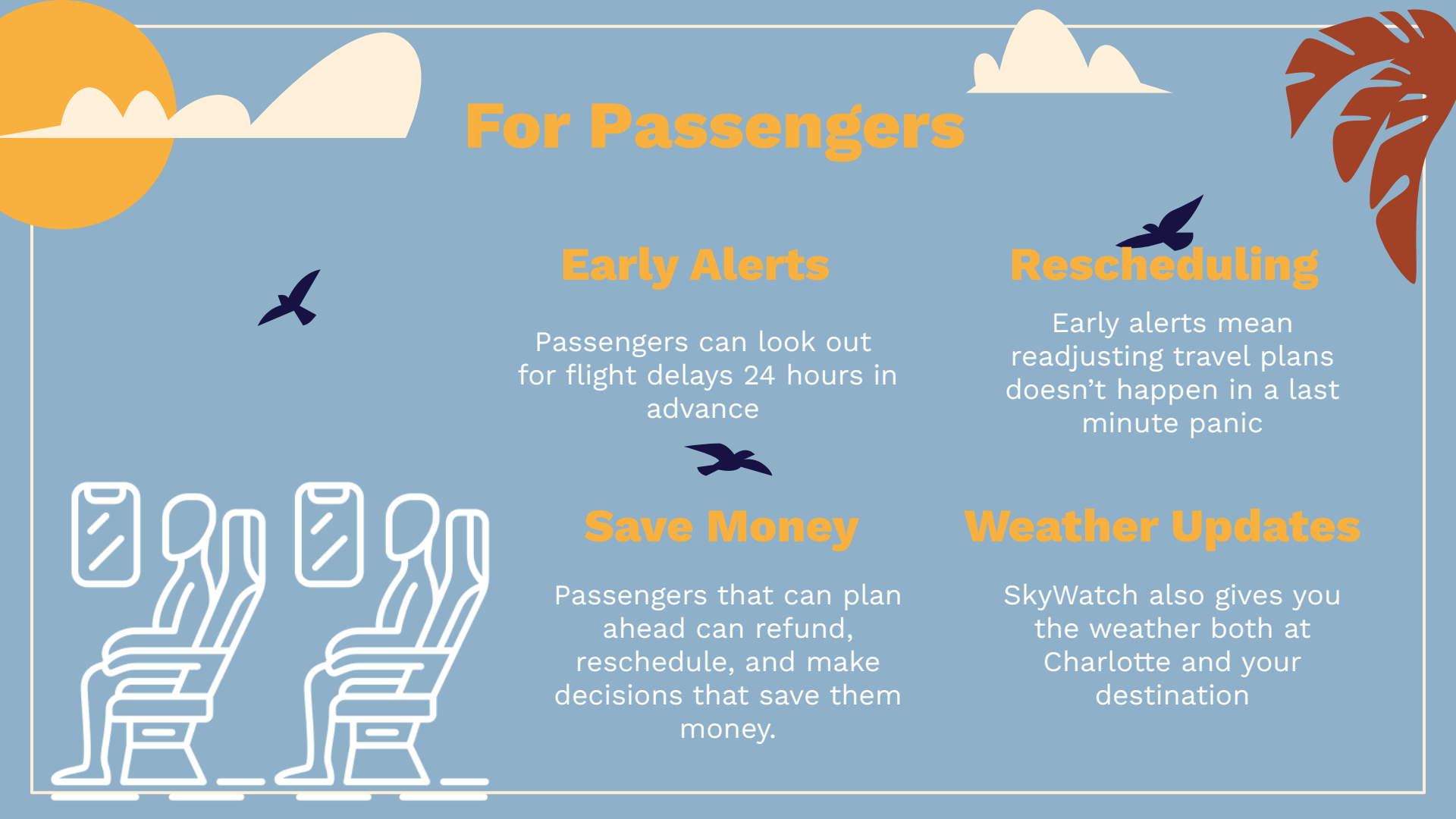
Flight

Select valid flight ▾

Predict Delay



**How could
this benefit
you?**



For Passengers

Early Alerts

Passengers can look out for flight delays 24 hours in advance

Rescheduling

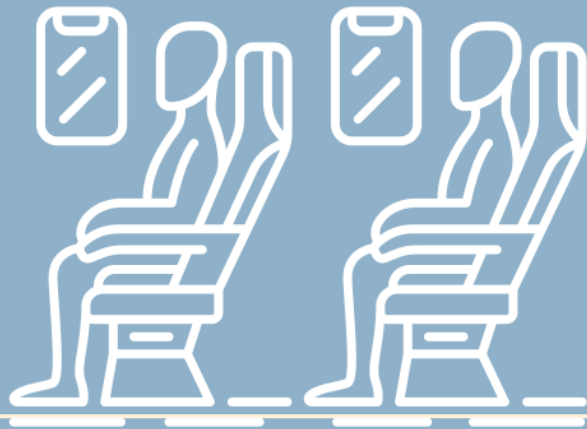
Early alerts mean readjusting travel plans doesn't happen in a last minute panic

Save Money

Passengers that can plan ahead can refund, reschedule, and make decisions that save them money.

Weather Updates

SkyWatch also gives you the weather both at Charlotte and your destination



For Airports

Scheduling

Additional, reliable data allows for effective flight scheduling that accounts for delays in advance

Customer Experience

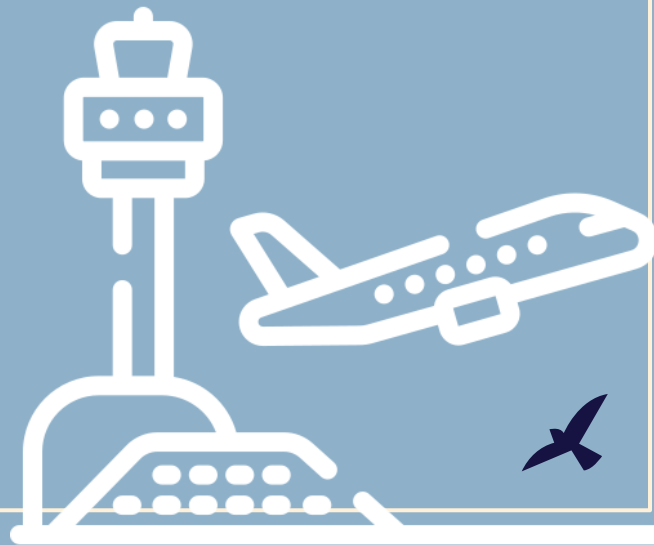
Allows customers a stress free and enjoyable experience

Redundancy

Traffic controllers additional metrics to rely on when making decisions

Customizable

Can easily be customized for any airport



Thanks!

Do you have any questions?



https://github.com/sania-k/ITC_S6190_CourseProject.git

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